

BharatPe: Scaling data effortlessly for analytics and AI with Google Cloud to promote digital payments

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BharatPe improved to promote digital payments among merchants by using managed BigQuery Vertex AI to scale learning pipelines with minimal infrastructure management.

About BharatPe

Founded in 2018 with the vision to make financial inclusion a reality for Indian merchants, BharatPe makes digital payments more accessible for over 10 million small offline merchants and [kirana store owners in the country](#).

(<https://www.dictionary.com/browse/kirana>), with its interoperable UPI QR which was the first in India with zero transaction fees. The company also facilitates loans to offline merchants in partnership with NBFCs, and launched India's first zero rental POS machine for merchants in

Google Cloud results

- Processes up to 80TB of data each day on BigQuery with minimal infrastructure management
- Supports a multi-cloud strategy with Cloud Composer to orchestrate more than 1,000 data workflows between clouds
- Reduced time to insight and enabled automated decisioning using AI/ML for critical business processes like fraud detection

General insight critical decision sector

2020. BharatPe also offers consumer fintech products like postpe. Named as the 'Fintech Company of the Year' at India Banking Summit & Awards 2022, BharatPe serves a network of 10 million merchants across 300+ cities in India.

Industries: Financial Services & Insurance

Location: India

(<https://cloud.google.com/composer/>), [Cloud Google Cloud \(<https://cloud.google.com/>\) aflow](https://cloud.google.com/dataflow/) (<https://cloud.google.com/dataflow/>) [Cloud BigQuery \(<https://cloud.google.com/bigquery/>\)](https://cloud.google.com/bigquery/)

Cloud Composer

(<https://cloud.google.com/composer/>)

Cloud SQL (<https://cloud.google.com/sql/>) x-ai/)

Dataflow (<https://cloud.google.com/dataflow/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Let us take on your challenge. We're here to help.

Document AI

(<https://cloud.google.com/document-ai/>)

Vision AI (<https://cloud.google.com/vision/>)

Vertex AI (<https://cloud.google.com/vertex-ai/>)

and credit decisions

Consumers in every Indian neighborhood are familiar with the ubiquitous grocery stores known as kiranas. These merchants sell everyday goods from curry powder to detergent on a cash and carry basis. As consumers move to cashless payments, kiranas will need to adapt. Fintech company BharatPe (<https://bharatpe.com/>) offers a QR code solution that helps millions of such offline merchants to accept cashless payments across a range of payment apps, without transaction fees.

Like a thumbprint, the BharatPe QR code is a unique identifier that contains the merchant's Unified Payments Interface (UPI) ID, linking digital payments with their designated bank account. Customers pay for their purchases by scanning the BharatPe QR code app with their preferred UPI payment app. Given the massive amounts of data generated daily, from payment processing to business analysis, the company aggregates the raw data into usable insights on Google Cloud (<https://cloud.google.com/>) for real-time decision-making.

"BigQuery (<https://cloud.google.com/bigquery>) is totally serverless so we don't need to take care of the infra level or check if we have enough clusters every time the data grows," says Sanjeev Singh, Head of Data Platform and Engineering at BharatPe. "This fully managed architecture along with ML and scheduling capabilities gives us more time to focus on writing business logic and implementing new use cases."

Singh further shares that with BigQuery capabilities, BharatPe is able to query several terabytes of data within minutes, given its capacity to concurrently run 100 queries by default, along with cache capabilities. BigQuery scheduling queries feature saves time and effort to write code, which previously took two to three days to complete on the older system.

The company ingests and distributes 80TB of data with Pub/Sub (<https://cloud.google.com/pubsub>) and orchestrates analytic and data science workflows in a multi-cloud environment with Cloud Composer (<https://cloud.google.com/composer>) that integrates with BigQuery (<https://cloud.google.com/bigquery>) and Dataflow (<https://cloud.google.com/dataflow>). Document AI (<https://cloud.google.com/document-ai>) and Vision AI (<https://cloud.google.com/vision>) are used to extract data and insights from onboarding documents to automate its Know-Your-Customer (KYC) process. Additionally, with the autoscale capabilities of DataProc, Dataflow, Pub/Sub, Cloud Function and Cloud Storage, without user intervention, the data science team was able to free up time to focus more on writing business logic, ideation and building new data products.

"With BigQuery, the data science team prepares data in a way that business users can understand and find KPIs that matter," says Sunil Saini, Head of Data Sciences at BharatPe.

"Through our collaboration with Google Cloud, we are able to leverage our data capability to fulfill the financial needs of our merchants and consumers. Using Google Data and AI services, we have been able to tap into more consumer moments and ultimately drive BharatPe ahead of the Indian fintech market."

—Vijay Kumar Aggarwal,
Chief Technology Officer
(CTO), BharatPe

Processing data at speed and scale for real-time decision making

Prior to Google Cloud, BharatPe managed its legacy data warehouse with limited capacity to run a large number of queries. The company ran key performance indicator (KPI) reports, without the ability to understand real-time data patterns.

"Speed matters specifically in our world where we implement changes at a very fast pace," shares Singh. "When we introduce a process or add a feature, we need to constantly measure and analyze the impact of that change and intervene quickly if something is going wrong."

Singh agrees, and shares that BharatPe operates in a multi-cloud environment for disaster recovery and leverages best-of-class technology. Hence, it also needs a data platform that can run queries against data, regardless of where it resides. "BigQuery provides the ideal solution to support our multi-cloud strategy with significant improvements in query performance."

Loading three months of data for quarterly reports took more than 30 minutes on the legacy system, and in some cases, queries failed because the system could not scale to support analytical needs. On BigQuery, BharatPe can run queries on three

years of data and get the required answers within a few seconds.

Using Cloud Composer and Dataflow, BharatPe can create data pipelines from any source in its BigQuery data warehouse. This means the company is able to scale its data processing, especially for data-intensive workflows such as machine learning. BharatPe currently runs more than 1,000 Directed Acyclic Graphs (DAGs) in its Cloud Composer environment.

Singh adds, "Google Cloud helps us process data at scale and speed. We utilize a familiar SQL interface in BigQuery to manage data processing and feature engineering pipelines. The serverless architecture allows us to manage pipelines with a lean team."

"Running our Data platform on Google BigQuery helps us improve our consumers' experience and keeps them engaged on our platform. I look forward to using

Google Data and
AI services to
innovate and
create even
more successful
cases in
BharatPe."

—Geetanshu Singla, Vice
President of Engineering,
BharatPe

Improving credit access to small merchants with data insights

Beyond payment systems, loan financing is also an integral part of BharatPe's business. For kiranas and small offline merchants, the lack of credit history makes it difficult to get loans, be it for purchasing inventory or to cover unexpected costs. BharatPe wants to improve credit access for these merchants and in turn, help them grow their business.

Using machine learning, BharatPe speeds up loan approval decisions with minimal manual intervention from loan officers. To apply for a loan, merchants must submit KYC information, including their Permanent Account Number (PAN) card, proof of address, and relevant images such as a storefront image and stock kept in the store. BharatPe matches names with FuzzyWuzzy logic in BigQuery to overcome spelling variations.

With the help of VisionAI, BharatPe detects and extracts the store name and address from photos and compares that with the information in its database and the borrower's loan application. This verification minimizes identity fraud, where the borrower pretends to be a legitimate store owner but funnels the loan into a non-related bank account. By automating these processes, the turnaround time has decreased from two days to just five seconds.

BharatPe examines the merchant's loan profile using a machine learning algorithm that analyzes variables such as transaction volume and frequency to determine a merchant's borrowing capacity. The more BharatPe QR code-based payments the merchant accepts, the bigger the loan amount. Machine learning model outcomes are directly integrated into the business decision processes as a microservice and enable real-time decisions.

By analyzing transaction data in BigQuery, BharatPe creates a loan profile on each applicant that determines the loan amount. Some merchants get their loans approved almost immediately to start using the loan amount.

Likewise, BharatPe also uses machine learning for KYC and underwriting for its consumer-facing postpe app. By analyzing patterns in the customer's payment behavior, the company helps them get instant credit for buy-now-pay-later (BNPL) purchases, without the hassle of applying for a credit card. The seamless checkout process helps BharatPe attract a large consumer market. postpe is

one of the fastest growing BNPL products in India today.

"Leveraging the data platform on Google Cloud helps us to capture and analyze data in real-time for decision-making across the business. Along with ML capabilities, BigQuery streaming data is a major advantage for us."

—Sanjeev Singh, Head of
Data Platform and
Engineering, BharatPe

Uncovering marketing insights on BigQuery to attract customers

The marketing team at BharatPe is also using data to identify challenges and make timely decisions. For example, by analyzing campaign data to attract new customers, the team can refine its acquisition strategy by answering questions such as, "Which cities drive the maximum number of downloads?" and "What channels do customers use to find the app?"

Apart from coupons on ecommerce and grocery websites to attract new customers, BharatPe uses in-app push notifications to increase engagement from existing customers. The company receives millions of data points per second from its digital marketing partners such as AppsFlyer for marketing analytics, and Clevertap for in-app message analytics. BharatPe integrates the marketing data into Pub/Sub that's capable of sending more than 500 million messages per second.

"BigQuery is our Single Source of Truth that gathers data from different sources. We compute more than 200 attributes, regarding merchant's transactional, behavioral and demographics profile," says Saini. "This data increases the efficiency of digital marketing campaigns such as retention and product lifecycle journeys."

By recognizing signals, BharatPe automatically triggers the appropriate action to address the customer's pain points throughout their journey. For example, sending a push notification to an inactive

customer to find out why he is not using the app may help to retain the customer.

"Google Cloud has helped us achieve positive outcomes through use cases such as credit risks and real-time fraud identification. Moving forward, we plan to use Google Cloud for model lifecycle management so as to accelerate the use of AI for innovative financial services that our customers need and want."

*—Sunil Saini, Head of
Data Science, BharatPe*

Accelerating AI for next-generation financial services

According to Saini, BharatPe currently uses Vertex AI (<https://cloud.google.com/vertex-ai>) as its data science workbench that tightly integrates with BigQuery to train its predictive models, compute how it should be deployed, and run the online prediction.

"Google Cloud has helped us achieve positive outcomes through use cases such as credit risks and real-time fraud identification," says Saini.

"Moving forward, we plan to use Google Cloud for model lifecycle management to accelerate the use of AI for innovative financial services that our customers need and want."